

Practical exercises

— area of triangles and quadrilaterals

A Cambridge insert: one height, one base, and a compound shape cut into pieces you already know.

LESSON 94–95

2 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 15

LESSON CLOCK

10'

25'

25'

20'

5'

The four formulae	10 min
Choosing the right height	25 min
Compound shapes	25 min
Units and estimation	20 min
Homework	5 min

LEARNING OBJECTIVES

- Use the area formulae for a rectangle, a triangle, a parallelogram and a trapezium.
- Identify the height that belongs to a chosen base.
- Find the area of a compound shape by cutting or by subtracting.
- Work with the units of area and convert between them.

TEXTBOOK

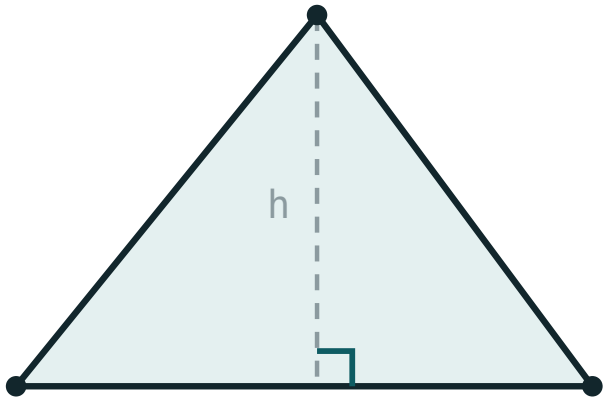
National	pp. 277–281
Cambridge	Stage 8, pp. 148–156
Workbook	Exercise 15.2

01

Explanation

The four formulae

$$S = ab \quad S = \frac{1}{2}ah \quad S = ah \quad S = \frac{a + b}{2} \cdot h$$



a

$$S = \frac{1}{2} \cdot a \cdot h$$

Base and perpendicular height

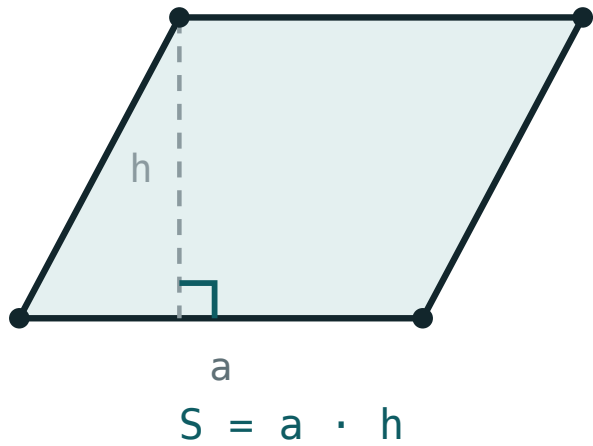
SHAPE	FORMULA	WHAT <i>h</i> MEANS
rectangle	ab	the second side
triangle	$\frac{1}{2}ah$	perpendicular from the base to the opposite vertex
parallelogram	ah	perpendicular between the two parallel sides
trapezium	$\frac{a+b}{2} \cdot h$	perpendicular between the parallel sides

THREE FORMULAE, ONE IDEA

A parallelogram is two copies of a triangle, and a trapezium is a parallelogram whose two parallel sides have been averaged. Learning $S = \frac{1}{2}ah$ and the reason for it makes the other two memorable rather than memorised.

Choosing the right height

The height must be perpendicular to **the base you chose**. A slanting side is never a height, and in an obtuse triangle the height may fall outside the figure.



The height is perpendicular, not slanting

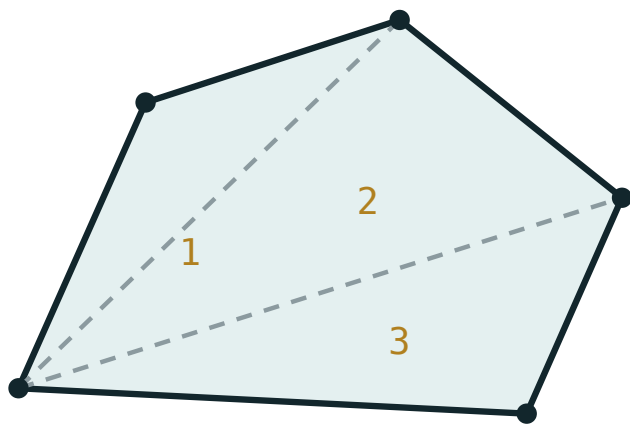
FIGURE	BASE	HEIGHT	AREA
triangle	10 cm	6 cm	30 cm ²
parallelogram	8 cm	5 cm	40 cm ²
trapezium	6 cm and 10 cm	4 cm	32 cm ²
rectangle	7 cm	3 cm	21 cm ²

THE SLANTING SIDE IS NOT THE HEIGHT

A parallelogram with sides 8 and 6 and height 5 has area 40, not 48. Using the second side instead of the perpendicular is the single commonest area error in the paper.

Compound shapes

Two routes work for any compound figure: cut it into known pieces and add, or enclose it in a rectangle and subtract what is not wanted.



split into triangles, then add the areas

Cutting a compound shape into known pieces

SHAPE	METHOD	WORKING	AREA
an L-shape 10×8 with a 4×3 corner removed	subtract	$80 - 12$	68 cm^2
a house outline: 6×4 rectangle under a triangle of height 3	add	$24 + 9$	33 cm^2
a path 1 m wide round a 5×4 plot	subtract	$42 - 20$	22 m^2

LABEL THE PIECES BEFORE YOU CALCULATE

Write the area of each piece next to it on the sketch. Nearly every lost mark on a compound shape is a piece counted twice or left out, not an arithmetic slip.

Units and estimation

CONVERSION	FACTOR	WHY
$1 \text{ cm}^2 = 100 \text{ mm}^2$	10^2	both sides scaled by 10
$1 \text{ m}^2 = 10\,000 \text{ cm}^2$	100^2	both sides scaled by 100
$1 \text{ ha} = 10\,000 \text{ m}^2$	100×100	a square 100 m on a side
$1 \text{ km}^2 = 100 \text{ ha}$	1000^2	a square 1 km on a side

LENGTHS SCALE BY k , AREAS BY k^2

A metre is *100* centimetres, but a square metre is *10 000* square centimetres. Using *100* for the area conversion is wrong by a factor of a hundred.

02 Worked examples

EXAMPLE 1 A triangle has base *10 cm* and height *6 cm*. Find its area.

$$① \quad S = \frac{1}{2}ah$$

$$② \quad S = \frac{1}{2} \cdot 10 \cdot 6$$

$$③ \quad S = 30 \text{ cm}^2$$

ANSWER

30 cm²

EXAMPLE 2 A trapezium has parallel sides *6 cm* and *10 cm* and height *4 cm*. Find its area.

$$① \quad S = \frac{a+b}{2} \cdot h$$

$$② \quad S = \frac{6+10}{2} \cdot 4$$

$$③ \quad S = 8 \cdot 4$$

$$④ \quad S = 32 \text{ cm}^2$$

ANSWER

32 cm²

EXAMPLE 3 A rectangular plot 5 m by 4 m has a path 1 m wide all round it. Find the area of the path.

- ① The outer rectangle is 7 m by 6 m.
One metre on each side.

② $7 \cdot 6 = 42 \text{ m}^2$

③ $5 \cdot 4 = 20 \text{ m}^2$

④ $42 - 20 = 22 \text{ m}^2$

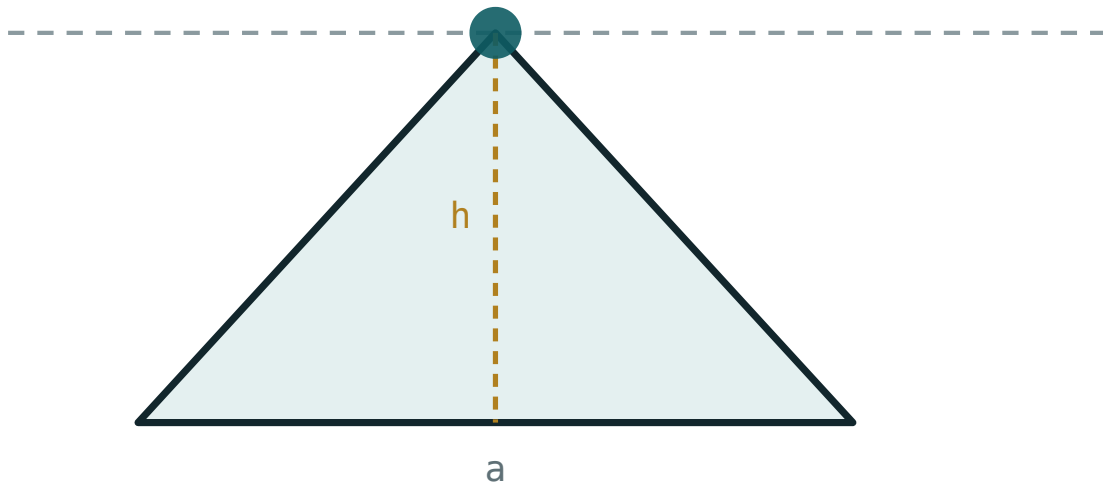
ANSWER

22 m^2

03 Interactive model

Give one parallelogram with all three lengths marked — base, slanting side and height — and ask for the area; the marks separate those who read the figure from those who multiply the first two numbers.

Slide the apex — the area does not move

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Area	Yuza	Площадь
Base	Asos	Основание
Height	Balandlik	Высота
Perpendicular height	Perpendikulyar balandlik	Перпендикулярная высота
Compound shape	Murakkab shakl	Составная фигура
Trapezium	Trapetsiya	Трапеция
Square centimetre	Kvadrat santimetr	Квадратный сантиметр
Hectare	Gektar	Гектар

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The area of a triangle is:

The area of a parallelogram is:

In the formulae, h must be:

A trapezium with parallel sides 6 and 10 and height 4 has area:

1 m² equals:

1 ha equals:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Triangle, base 10, height 6

ANSWER 30 cm^2

2. Rectangle 7×3

ANSWER 21 cm^2

3. Parallelogram, base 8, height 5

ANSWER 40 cm^2

4. Trapezium 6 and 10, height 4

ANSWER 32 cm^2

5. Triangle, base 12, height 5

ANSWER 30 cm^2

6. 1 cm^2 in mm^2

ANSWER 100

7. 1 m^2 in cm^2

ANSWER 10 000

Medium · the standard question

7 PROBLEMS

1. Parallelogram with sides 8 and 6 and height 5

ANSWER 40 cm^2

2. A triangle of area 24 has base 8: find the height

ANSWER 6

3. A trapezium of area 45, parallel sides 6 and 9: find the height

ANSWER 6

4. L-shape: 10×8 with a 4×3 corner removed

ANSWER 68 cm^2

5. House outline: 6×4 rectangle plus a triangle of height 3

ANSWER 33 cm^2

6. A 1 m path round a 5×4 plot

ANSWER 22 m^2

7. 3 ha in m^2

ANSWER 30 000

Hard · stretch and reason

7 PROBLEMS

1. A rhombus with diagonals 10 and 12

ANSWER 60 cm^2

2. A triangle with base a and height h has its base doubled: the new area

ANSWER *twice the old*

3. Both dimensions of a rectangle are trebled: the area is multiplied by

ANSWER 9

4. A field of 2.5 ha in m^2

ANSWER $25\,000$

5. A trapezium has parallel sides x and $x + 6$ and height 4 : its area

ANSWER $4x + 12$

6. Two triangles have the same base and lie between the same parallels: their areas are

ANSWER *equal*

7. A 12×9 rectangle contains a triangle on the 12 side with apex on the opposite side: its area

ANSWER 54 cm^2

07 Homework

Homework — 5 tasks

Sketch every shape and write the area of each piece beside it.

- Find the area of a triangle with base 14 cm and height 9 cm .
- A parallelogram has sides 12 and 7 and height 5 . Find its area and say which number is not used.
- Find the area of a trapezium with parallel sides 8 and 14 and height 6 .
- A 2 m path runs round a 10 m by 6 m garden. Find the area of the path.

5. Convert 4.5 ha to square metres and $250\,000\text{ cm}^2$ to square metres.